LUKE MAY

Guelph, Ontario · luke.may1181@gmail.com · Professional portfolio edition, October 2026

PROFESSIONAL SUMMARY

Mathematical Sciences MSc graduate with research experience in differential-equation modelling, agent-based simulation, scenario analysis, and quantitative problem-solving. Skilled in Python, R, numerical methods, and communicating technical findings. Experienced in teaching and explaining complex mathematics to varied audiences.

EDUCATION

Master's Degree in Mathematical Sciences 2026
University of Guelph · 4.0 GPA · Entrance bursary and scholarships totalling $15,666

Bachelor's Degree in Mathematical Sciences 2023
University of Guelph · 4.0 GPA · $3,000 entrance scholarship

RESEARCH AND SELECTED PROJECTS

MSc thesis · Modeling Expected Health of Different Ages Around the World in Relation to Environmental Factors

- Developed a differential-equation-based, age-structured population model encompassing children, youth, adults, and seniors across six countries.
- Investigated environmental conditions and scenarios including drought, air quality, food availability, and disease-related shocks over a 65-year modelling horizon.
- Performed stochastic simulation and scenario analysis and linked model outcomes with a composite health index.

Agent-Based Modelling of Public Transit Adoption · Fourth-year team project, 2022–2023

- Collaborated in a three-person research team to model incentives for public transportation adoption.
- Represented individual decisions and behavioural responses to different policy scenarios and examined system-level transit uptake.

TECHNICAL SKILLS

Programming & tools: Python, R, LaTeX, Microsoft Excel and Office, Google Workspace.

Methods: Mathematical and statistical modelling, differential equations, agent-based modelling, numerical methods, simulation and scenario analysis, quantitative data analysis, technical writing, research communication.

TEACHING AND COMMUNICATION EXPERIENCE

Teaching Assistant Sep 2023–Dec 2025
University of Guelph, Guelph, ON

- Supported instruction in seven undergraduate mathematics courses, including Calculus I and II, Linear Algebra, Numerical Methods, and Business Mathematics.
- Led tutorials, graded work, and provided individual academic support, translating challenging quantitative concepts into clear explanations.

Mathematics Tutor Sep 2021–Present
Oxford Learning, Guelph, ON

- Tutored students from Grade 6 through fourth-year university mathematics, adapting instruction to varied learning needs.
- Coordinated concurrent tutoring sessions for up to 15 students while maintaining clear organization and communication.

ADDITIONAL EXPERIENCE

Student Labourer May 2021–Aug 2022
EACOM, Sault Ste. Marie, ON

Worked independently and collaboratively in a high-volume lumber yard, adapting to operational priorities.

Tree Planter May–Jul 2020
Brinkman Reforestation, Northern Ontario

Planted 104,950 trees across 56 workdays (approximately 1,875 per day), the highest total on the team.